COURSE OUTCOMES

CO1	Understand the basic concepts of statistics and probability
CO2	Apply techniques of statistics and probability to solve engineering problems
CO3	Analyze engineering problems using statistics and probability
CO4	Develop mathematical solutions to various real time problems using MATLAB

Module – 1

PROBABILITY DISTRIBUTION

1. Define a) Random variable b) Discrete random variable c) continuous random variable

   ii) Discrete probability distribution with an example.

2. Suppose a random variable X takes the values: -3,-1, 2 and 5 with respective probabilities $\frac{2k-3}{10}, \frac{k-2}{10}, \frac{k-1}{10}, \frac{k+1}{10}$. Find the value of k and i) find P [-3<X<4] and ii) P[X $\leq$ 2].

3. Find the constant c, such that the function $f(x) = \begin{cases} cx^2, 0 < x < 3 \\ 0, otherwise \end{cases}$ is a p.d.f. Also compute p(1<x<2), p(x≤1), p(x>1).

4. The probability distribution of a finite random variable X is given by the following table:

X_i	-2	-1	0	1	2	3
$P(x_i)$	0.1	k	0.2	2k	0.3	k

   Determine the value of k and find the mean, variance and standard deviation.

5. The probability density function of a variate x is

x	0	1	2	3	4	5	6
P(x)	K	3K	5K	7K	9K	11K	13K

   Find K. Find P(x<4), P(x $\geq$ 5) and P(3<x $\leq$ 6).

6. A Random variable x has the following distributions:

x:	-2	-1	0	1	2	3	4
P(x)	0.1	0.1	k	0.1	2k	k	k

Find k, mean and S.D of the distribution.

7. The probability distribution of a finite random variable X is given by the following table:

x	0	1	2	3	4	5	6	7
P(x)	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find k, p(x<6), p(x≥6), p(3<x≤6)

8. Find the value of 'k' such that the following distribution represents a finite probability distribution. Hence find the mean (μ) and standard deviation (σ). Also find p(x≤1),p(x>1) and p(-1<x≤2).

x	-3	-2	-1	0	1	2	3
P(x)	k	2k	3k	4k	3k	2k	k

9. The probability density of a continuous random variable is given by $p(x)= y_0\, e^{-|x|}, -\infty < x < \infty$. Find y_0. Also find the mean.

10. The probability that a man aged 60 will live up to 70 is 0.65. What is the probability that out of 10 men, now aged 60, i) exactly 9, ii) atmost 9 iii) atleast 7, will live up to the age of 70 years.

11. The probability that an individual suffers a bad reaction from an injection is 0.001. Find the probability that out of 2000 individuals, more than 2 will get a bad reaction.

12. Given that 2% of the fuses manufactured by a firm are defective, find by using Poisson distribution, the probability that a box containing 200 fuses has i) No defective fuses ii) 3 or more effective fuses iii) At least one defective fuse.

13. A die is tossed thrice. A success is getting 1 or 6 on a toss. Find the mean and variance of the number of successes.

14. Alpha – particles are emitted by a radioactive source at an average of 5 emissions in 20 minutes. What is the probability that there will be i) exactly two emissions ii) at least two emissions in 20 minutes?

15. The probability that a pen manufactured by a company will be defective is 0.1 if 12 such pens are selected. Find the probability that i) exactly 2 will be defective, ii) at least 2 will be defective, iii) none will be defective.

16. In a certain town the duration of shower has mean 5 minutes. What is the probability that shower will last for i) 10 minutes or more; ii) less than 10 minutes; iii) between 10 and 12 minutes.

17. In a quiz contest of answering 'YES' or 'NO', what is the probability of guessing at least 6 answers correctly out of 10 questions asked? Also find the probability of the same if there are 4 options for a correct answer?

18. The probability that a bomb dropped hits the target is 0.002. Find the probability that out of 600 bombs dropped i) Exactly 2 will hit the target ii) At least 3 will hit the target.

19. The life of an electric bulb is normally distributed with average life of 2000 hours and standard deviation of 60hours. Out of 2500 such bulbs, find the number of bulbs that are likely to last between 1900 and 2100 hours. Given $[0 \leq z \leq 1.67]$ =0.4525.

20. In a test on 2000 electric bulbs, it was found that the life of a particular make was normally distributed with an average life of 2040 hours and S.D. of 60 hours. Estimate the number of bulbs likely to burn for a) More than 2150 hours b) Less than 1950 hours and c) More than 1920 hours, but less than 2160 hours.

21. The mark of 100 students in an examination follows a normal distribution with mean 70 and standard deviation 5. Find the number of students whose marks will be i) less than 65 ii) more than 75 iii) Between 65 and 75.

22. If the mean and standard deviation of the number of correctly answered questions in a test given to 4096 students are 2.5 and $\sqrt{1.875}$. Find an estimate of the number of candidates answering correctly (i) 8 or more questions (ii) 2or less (iii) 5 questions.

23. In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation, given that A (0.5) =0.19 and A (1.4) =0.42.

24. In an examination 7% of students' scores less than 35% marks and 89% of students score less than 60% marks. Find the mean and standard deviation of the marks are normally distribute, it is given that $P(0<z<1.2263) = 0.39$ and $P(0<z<1.4757) = 0.43$.

25. Suppose the life time of certain kind of energy backup battery (in Hrs)is a random variable 'x' having Weibull distribution $\alpha=0.1$, $\beta=0.5$. Find the probability that such a battery will last for more than 300 hours.

26. Suppose the time to failure (in min) of certain electric components subjected to continuous vibrations may be looked upon has a variable Weibull distribution α=0.33, β=0.2. (1) How long such a component is expected to last. (2) What is the probability that such a component will fail in less than 5 hours.

27. A random variable X has a uniform distribution over (-3,3), find k for which $P(X > k) = \frac{1}{3}$

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Module – 2

JOINT PROBABILITY DISTRIBUTION AND STATISTICS

1. The joint distribution of two variables X and Y are given below:

X \ Y	-4	2	7
1	1/8	1/4	1/8
5	1/4	1/8	1/8

Compute the following: (i) E(X) and E(Y) (ii) E(XY) (iii) COV(X,Y) **(7M)(JUL-07, DEC-18)**

2. The joint distribution of two variables X and Y are given below:

X \ Y	2	3	4
1	0.06	0.15	0.09
2	0.14	0.35	0.21

Determine Marginal distributions of X and Y. Also verify that X and Y independent.

(7M,JAN-05,09,JUL-04)

3. The joint distribution of two variables X and Y are given below:

X \ Y	1	3	6
1	1/9	1/6	1/18
3	1/6	1/4	1/12
6	1/18	1/12	1/36

Determine Marginal distributions of X and Y. Also verify that X and Y independent. **(7M)(JAN-06)**

4. The joint distribution of two variables X and Y are given below:

X \ Y	1	3	9

2	1/8	1/24	1/12
4	1/4	1/4	0
6	1/8	1/24	1/12

Determine Marginal distributions of X and Y and evaluate COV(XY). **(7M)(DEC-18, JUL-09,11)**

5. The joint distribution of two variables X and Y are given below:

X \ Y	-3	2	4
1	0.1	0.2	0.2
3	0.3	0.1	0.1

Determine (i)Marginal distributions of X and Y(ii) Covariance of X and Y. **(7M)(JAN-09,JUL-05)**

6. The joint probability distribution of two random variables X and Y are given below:

X \ Y	-2	-1	4	5
1	0.1	0.2	0	0.3
2	0.2	0.1	0.1	0

Find the Marginal distributions of X ,Y. Also find the Covariance of X and Y. **(7M)(JUL-08,DEC-10,19)**

7. Compute (i) $P(x = 1, y = 2)(ii)P(x \geq 1, y \leq 2)(iii)P(x \leq 1, y \leq 2)(iv)P(x + y \geq 2)$, using the following joint probability distribution for x and y. **(7M)(DEC-11)**

x \ y	0	1	2	3	Sum
0	0	1/8	1/4	1/8	1/2
1	1/8	1/4	1/8	0	1/2
Sum	1/8	3/8	3/8	1/8	1

8. X and Y are independent random variables, X take the values 1,2 with probability 0.7; 0.3 and y take the values -2, 5, 8 with probabilities 0.3, 0.5, 0.2. Find the joint distribution of X and Y hence find cov(X,Y). **(6M)(JAN-18)**

9. The joint distribution of two discrete variables x and y is f(x,y)=k(2x+y) where x and y are integers such that $0 \leq x \leq 2$; $0 \leq y \leq 3$. Find (i) the value of k; (ii) Marginal distribution of x and y; (iii) Are x and y independent? **(6M)(JAN-18)**

10. With usual notation compute $\bar{x}$, $\bar{y}$ and r from the following lines of regression $2x + 3y + 1 = 0$ and $x + 6y - 4 = 0$.

11. The lines of regressions are $3x + 2y = 26$ and $6x + y = 31$. Find i) means of the variables x and y ii) Correlation coefficient between x and y.

12. While calculating correlation coefficient between two variables x and y from 25 pairs of observations, the following results were obtained:

$$n = 25, \quad \sum x = 125, \sum x^2 = 650, \sum y = 100, \sum y^2 = 460, \sum xy = 508.$$

Later it was discovered at the time of checking that the pairs of values (8,12) and (6,8) were copied down as (6,14) and (8,6). Obtain the correct value of correlation coefficient.

13. Find the coefficient of correlation for the following data:

x	55	56	58	59	60	60	62
y	35	38	39	38	44	43	45

14. Obtain the coefficient of correlation and the lines of regression

x	1	3	4	2	5	8	9	10	13	15
y	8	6	10	8	12	16	16	10	32	32

15. The two regression equations of the variables x and y are $x = 19.13 - 0.87y$ and

$y = 11.64 - 0.50x$ Find i) mean of x's ii) mean of y's and

iii) the correlation coefficient of x and y.

16. Find the coefficient of correlation, line of regression x on y and the line of regression of y on x; given

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

17. In a partially destroyed lab record of analysis of correlation data, the following results are available. Variance of x is 9. Regression equations are $8x - 10y + 66 = 0$ and $40x - 18y - 214 = 0$. Find $\bar{x}, \bar{y}, \sigma_y$ and correlation coefficient.

18. The lines of regression of y on x and x on y are given as $4x - 5y + 33 = 0$ and

$20x - 9y = 107$ respectively. Calculate means of x and y and coefficient of correlation.

19. Obtain the lines of regression and hence find coefficient of correlation for the following data

x	1	3	4	2	5	8	9	10	13	15
y	8	6	10	8	12	16	16	10	32	32

20. Find the Coefficient of correlation and regression lines for the following data

x	1	2	3	4	5	6	7	8	9	10
y	10	12	16	28	25	36	41	49	40	50

21. Calculate the coefficient of correlation for the following ages of husbands and wives

Husband's age: x	23	27	28	28	29	30	31	33	35	36
Wife's age: y	18	20	22	27	21	29	27	29	28	29

22. Calculate the rank coefficient of correlation for 10 students who have obtained the following percentage of marks in Mathematics and Electronics.

Marks in Mathematics	78	36	98	25	75	82	90	62	65	39
Marks in Electronics	84	51	91	60	68	62	86	58	53	47

22. Calculate the Karl's Pearson rank coefficient for the following data

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70